

12.421/12.621 Lecture 6: Reflection and Transmission of EM Waves at Oblique Incidence and Multiple Layers

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Topics covered

- Review: EM boundary conditions and the laws of geometric optics
- Reflection from a thin layer overlying a half space (two-layer composite)
- Multiple reflections in a layer

1 Lecture 5 review

1.1 EM boundary conditions

The EM boundary conditions (BCs) for zero charge and zero current at the interface, in lossless media (Lectures 4 and 5), with $\hat{\mathbf{a}}$ the unit normal to the interface, are:

1. $\varepsilon'_1 E_1^\perp = \varepsilon'_2 E_2^\perp$ or $\hat{\mathbf{a}} \cdot (\varepsilon'_1 \mathbf{E}_1 - \varepsilon'_2 \mathbf{E}_2) = 0 \rightarrow$ the normal component of $\varepsilon' \mathbf{E}$ is continuous;
2. $\mu_1 H_1^\perp = \mu_2 H_2^\perp$ or $\hat{\mathbf{a}} \cdot (\mu_1 \mathbf{H}_1 - \mu_2 \mathbf{H}_2) = 0 \rightarrow$ the normal component of $\mu \mathbf{H}$ is continuous;
3. $\mathbf{E}_1^\parallel = \mathbf{E}_2^\parallel$ or $\hat{\mathbf{a}} \times (\mathbf{E}_1 - \mathbf{E}_2) = \mathbf{0} \rightarrow$ the tangential component of $\mathbf{E}$ is continuous;
4. $\mathbf{H}_1^\parallel = \mathbf{H}_2^\parallel$ or $\hat{\mathbf{a}} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{0} \rightarrow$ the tangential component of $\mathbf{H}$ is continuous.

1.2 Laws of geometric optics

1. The incident, reflected, and transmitted rays and the normal to the interface are coplanar $\rightarrow$ plane of incidence.
2. $\theta_i = \theta_r = \theta_t$.
3. $\sqrt{\varepsilon_2} \sin \theta_2 = \sqrt{\varepsilon_1} \sin \theta_1$ (Snell's law).

2 Reflection from a thin layer ($d \lesssim \delta_p$) overlying a half space

- There are many cases where we may want to study thin-layer scattering, e.g., sea ice, snow on the ground, oil slicks on the ocean, etc.
- Consider the case of homogeneous materials with planar interfaces (Figure 1).
- If d is sufficiently small (defined later, but intuition ...), and
 - the reflectivity at interface 12 is sufficiently small and

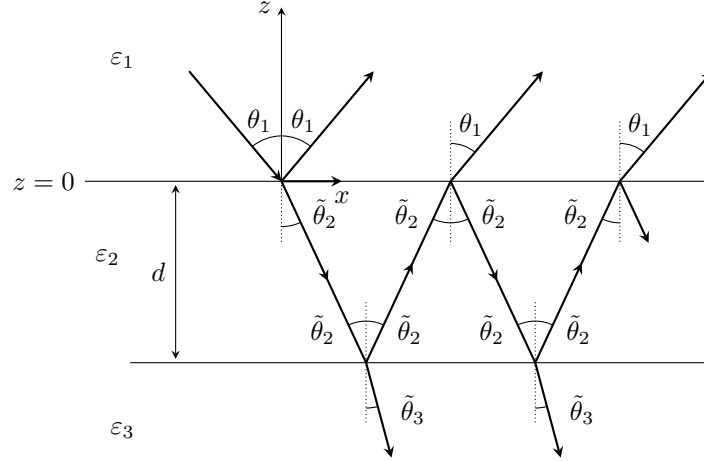


Figure 1: A layer (medium 2, ε_2) of thickness d between medium 1 (ε_1) and a half space (medium 3, ε_3). The incident ray is partly reflected at $z = 0$ and partly transmitted into the layer, where it is reflected back and forth between the two interfaces, transmitting into media 1 and 3 at each encounter.

- the reflectivity at interface 23 is sufficiently large,

then we can

- define an effective reflectivity of the two-layer composite in terms of the total upward-propagating $\mathbf{E}$ fields in medium 1 (through the effective reflection coefficient ρ_{eff} ; $\Gamma = |\rho_{\text{eff}}|^2$).
- Generally, we can define the reflectivity of an N -layer composite by defining the dielectric properties and applying the BCs at the $N - 1$ interfaces.
 - This can be constructed in a convenient matrix form for straightforward calculation of the effective reflectivity of the N -layer structure and the transmission into the bottom layer (the propagation matrix method; Ulaby & Long 2014, Sec. 2-10.2).
 - Here, we consider only the 3 layers pictured above (Figure 1).

2.1 Fields in medium 1

- Consider an H-polarized incident $\mathbf{E}$ field

$$\tilde{\mathbf{E}}_1^- = \hat{\mathbf{y}} A_1 e^{-\gamma_1 (x \sin \theta_1 - z \cos \theta_1)}, \quad (1)$$

where

- A_1 is the amplitude of the incident field (we called this E_h^i last lecture; simplified here);
- note that the superscript $-$ sign indicates downward propagation;
- note the use of numerical subscripts instead of $i, t, r \rightarrow$ to avoid confusion;
- note that $\gamma_1 \equiv j k_0 \sqrt{\varepsilon_1}$.
- The upward-propagating wave in medium 1 is

$$\tilde{\mathbf{E}}_1^+ = \hat{\mathbf{y}} B_1 e^{-\gamma_1 (x \sin \theta_1 + z \cos \theta_1)} = \hat{\mathbf{y}} \rho_{\text{eff}} A_1 e^{-\gamma_1 (x \sin \theta_1 + z \cos \theta_1)}, \quad (2)$$

where

- $B_1 \equiv$ amplitude of the sum of all upward $\mathbf{E}$ fields;
- note that the superscript $+$ indicates upward propagation;

$$\boxed{\rho_{\text{eff}} = \frac{B_1}{A_1} \equiv \text{effective reflection coefficient}} \quad (3)$$

- The total $\mathbf{E}$ field in medium 1 is

$$\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}_1^- + \tilde{\mathbf{E}}_1^+ = \hat{\mathbf{y}} A_1 \left(e^{\gamma_1 z \cos \theta_1} + \rho_{\text{eff}} e^{-\gamma_1 z \cos \theta_1} \right) e^{-\gamma_1 x \sin \theta_1}. \quad (4)$$

- We know that $\tilde{\mathbf{H}} = \frac{1}{\eta} (\hat{\mathbf{k}} \times \tilde{\mathbf{E}})$ (Lecture 2), applied to each of the downward- and upward-propagating waves

$\Rightarrow$ the total magnetic field in medium 1 is

$$\tilde{\mathbf{H}}_1 = \hat{\mathbf{x}} \tilde{H}_{1x} + \hat{\mathbf{z}} \tilde{H}_{1z}, \quad (5)$$

$$\tilde{H}_{1x} = \frac{A_1 \cos \theta_1}{\eta_1} \left(e^{\gamma_1 z \cos \theta_1} - \rho_{\text{eff}} e^{-\gamma_1 z \cos \theta_1} \right) e^{-\gamma_1 x \sin \theta_1}, \quad (6)$$

$$\tilde{H}_{1z} = \frac{A_1 \sin \theta_1}{\eta_1} \left(e^{\gamma_1 z \cos \theta_1} + \rho_{\text{eff}} e^{-\gamma_1 z \cos \theta_1} \right) e^{-\gamma_1 x \sin \theta_1}. \quad (7)$$

2.2 Fields in media 2 and 3

- In medium 2, we can follow the same procedure to get

$$\tilde{\mathbf{E}}_2 = \tilde{\mathbf{E}}_2^- + \tilde{\mathbf{E}}_2^+ = \hat{\mathbf{y}} \left(A_2 e^{\gamma_2 z \cos \tilde{\theta}_2} + B_2 e^{-\gamma_2 z \cos \tilde{\theta}_2} \right) e^{-\gamma_2 x \sin \tilde{\theta}_2}, \quad (8)$$

$$\tilde{\mathbf{H}}_2 = \hat{\mathbf{x}} \tilde{H}_{2x} + \hat{\mathbf{z}} \tilde{H}_{2z}, \quad (9)$$

$$\tilde{H}_{2x} = \frac{\cos \tilde{\theta}_2}{\eta_2} \left(A_2 e^{\gamma_2 z \cos \tilde{\theta}_2} - B_2 e^{-\gamma_2 z \cos \tilde{\theta}_2} \right) e^{-\gamma_2 x \sin \tilde{\theta}_2}, \quad (10)$$

$$\tilde{H}_{2z} = \frac{\sin \tilde{\theta}_2}{\eta_2} \left(A_2 e^{\gamma_2 z \cos \tilde{\theta}_2} + B_2 e^{-\gamma_2 z \cos \tilde{\theta}_2} \right) e^{-\gamma_2 x \sin \tilde{\theta}_2}. \quad (11)$$

- In medium 3, there is only downward propagation:

$$\tilde{\mathbf{E}}_3 = \hat{\mathbf{y}} \left(A_3 e^{\gamma_3 z \cos \tilde{\theta}_3} \right) e^{-\gamma_3 x \sin \tilde{\theta}_3}, \quad (12)$$

$$\tilde{\mathbf{H}}_3 = \hat{\mathbf{x}} \tilde{H}_{3x} + \hat{\mathbf{z}} \tilde{H}_{3z}, \quad (13)$$

$$\tilde{H}_{3x} = \frac{\cos \tilde{\theta}_3}{\eta_3} A_3 e^{\gamma_3 z \cos \tilde{\theta}_3} e^{-\gamma_3 x \sin \tilde{\theta}_3}, \quad (14)$$

$$\tilde{H}_{3z} = \frac{\sin \tilde{\theta}_3}{\eta_3} A_3 e^{\gamma_3 z \cos \tilde{\theta}_3} e^{-\gamma_3 x \sin \tilde{\theta}_3}. \quad (15)$$

- In lossy media, $\tilde{\theta}_2$ and $\tilde{\theta}_3$ are complex valued (cf. $\tilde{\theta}_t$ in Lecture 5).

2.3 Applying the BCs

- Tangential $\mathbf{E}$ field continuous:

$$\tilde{\mathbf{E}}_1|_{z=0} = \tilde{\mathbf{E}}_2|_{z=0} \Rightarrow (A_1 + B_1) e^{-\gamma_1 x \sin \theta_1} = (A_2 + B_2) e^{-\gamma_2 x \sin \tilde{\theta}_2}. \quad (16)$$

- Snell's law:

$$\gamma_1 \sin \theta_1 = \gamma_2 \sin \tilde{\theta}_2 = \gamma_3 \sin \tilde{\theta}_3 \quad (17)$$

$$\Rightarrow \boxed{A_1 + B_1 = A_2 + B_2} \quad (18)$$

- Tangential $\mathbf{H}$ field continuous:

$$\tilde{H}_{1x}|_{z=0} = \tilde{H}_{2x}|_{z=0} \Rightarrow \boxed{\frac{\cos \theta_1}{\eta_1} (A_1 - B_1) = \frac{\cos \tilde{\theta}_2}{\eta_2} (A_2 - B_2)} \quad (19)$$

- Similarly, at the 23 boundary,

$$\tilde{\mathbf{E}}_2|_{z=-d} = \tilde{\mathbf{E}}_3|_{z=-d} \quad \text{and} \quad \tilde{H}_{2x}|_{z=-d} = \tilde{H}_{3x}|_{z=-d} \quad (20)$$

$$\Rightarrow \boxed{\begin{aligned} A_2 e^{-\gamma_2 d \cos \tilde{\theta}_2} + B_2 e^{\gamma_2 d \cos \tilde{\theta}_2} &= A_3 e^{-\gamma_3 d \cos \tilde{\theta}_3} \\ \frac{\cos \tilde{\theta}_2}{\eta_2} (A_2 e^{-\gamma_2 d \cos \tilde{\theta}_2} - B_2 e^{\gamma_2 d \cos \tilde{\theta}_2}) &= \frac{A_3 \cos \tilde{\theta}_3}{\eta_3} e^{-\gamma_3 d \cos \tilde{\theta}_3} \end{aligned}} \quad (21)$$

2.4 Effective reflection coefficient of the two-layer composite

- Assuming A_1 is known, Eqs. (18)–(21) are 4 equations in 4 unknowns (B_1, A_2, B_2, A_3).
- Grinding through the algebra gives

$$\boxed{\rho_{\text{eff}} = \frac{\rho_{12} + \rho_{23} e^{-2\gamma_2 d \cos \tilde{\theta}_2}}{1 + \rho_{12} \rho_{23} e^{-2\gamma_2 d \cos \tilde{\theta}_2}}} \quad (22)$$

where the Fresnel reflection coefficients (Lecture 5) at the two interfaces are

$$\rho_{12} = \begin{cases} \frac{\eta_2 \cos \theta_1 - \eta_1 \cos \tilde{\theta}_2}{\eta_2 \cos \theta_1 + \eta_1 \cos \tilde{\theta}_2} & \text{H pol.,} \\ \frac{\eta_2 \cos \tilde{\theta}_2 - \eta_1 \cos \theta_1}{\eta_2 \cos \tilde{\theta}_2 + \eta_1 \cos \theta_1} & \text{V pol.,} \end{cases} \quad (23)$$

$$\rho_{23} = \begin{cases} \frac{\eta_3 \cos \tilde{\theta}_2 - \eta_2 \cos \tilde{\theta}_3}{\eta_3 \cos \tilde{\theta}_2 + \eta_2 \cos \tilde{\theta}_3} & \text{H pol.,} \\ \frac{\eta_3 \cos \tilde{\theta}_3 - \eta_2 \cos \tilde{\theta}_2}{\eta_3 \cos \tilde{\theta}_3 + \eta_2 \cos \tilde{\theta}_2} & \text{V pol.} \end{cases} \quad (24)$$

These have the forms of ρ_h and ρ_v in Lecture 5, with media $1 \rightarrow 2$ (ρ_{12}) and $2 \rightarrow 3$ (ρ_{23}).

3 A more direct approach: multiple reflections

But, now that we've laid the groundwork, let's take a more direct approach (dropping the x component of the phase because Snell's law tells us that the x values are constant in all media); see Figure 2.

3.1 Summing the multiple reflections

- Now, summing all upward-going waves in medium 1,

$$\begin{aligned}\rho_{\text{eff}} &= \rho_{12} + \tau_{21}\rho_{23}\mathcal{L}^2\tau_{12} + \tau_{21}\rho_{21}\rho_{23}^2\mathcal{L}^4\tau_{12} + \cdots \\ &= \rho_{12} + \tau_{12}\tau_{21}\rho_{23}\mathcal{L}^2 \sum_{p=0}^{\infty} \chi^p,\end{aligned}\tag{25}$$

where

$$\chi = \rho_{21}\rho_{23}\mathcal{L}^2.\tag{26}$$

- Note that $|\rho_{21}| < 1$, $|\rho_{23}| < 1$, and $|\mathcal{L}^2| \leq 1$ (with equality for a lossless layer)
 $\Rightarrow |\chi| < 1$ (for medium 1 air, media with $\epsilon' \geq 1$, and $\theta_1 < 90^\circ$).
- Also,

$$\rho_{21} = -\rho_{12} \quad \rightarrow \quad \tau_{21} = 1 + \rho_{21} = 1 - \rho_{12}.\tag{27}$$

Recall that

$$\tau_{12} = 1 + \rho_{12}.\tag{28}$$

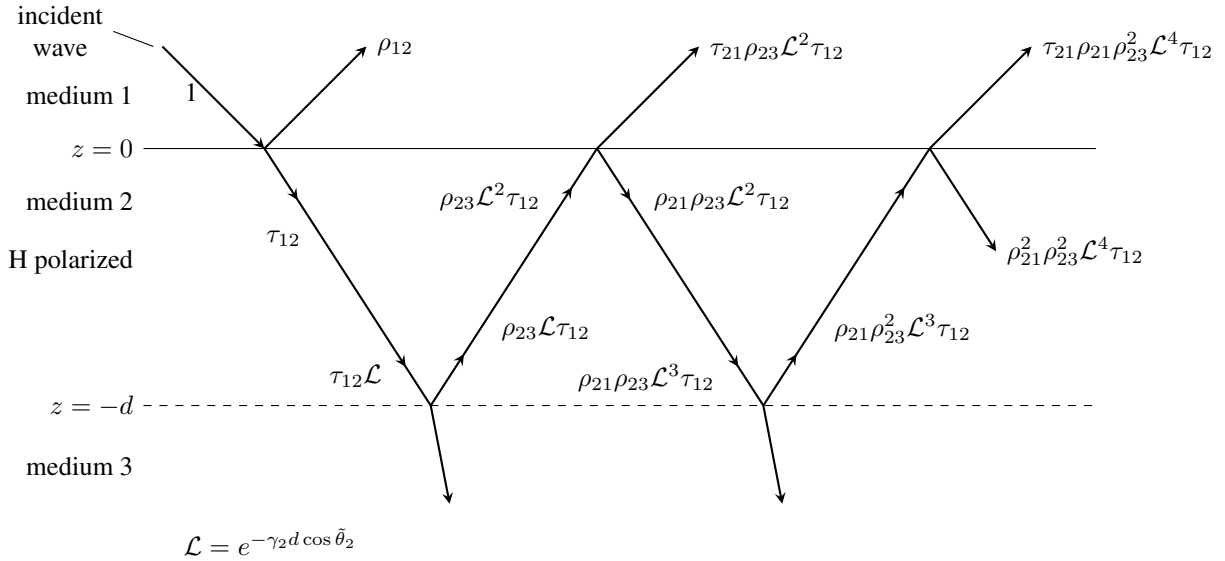


Figure 2: Multiple reflections of an H-polarized wave of unit amplitude incident from medium 1 on the layer (medium 2). Each label gives the amplitude of the wave at that point (the x component of the phase is dropped). ρ_{12} and τ_{12} are the reflection and transmission coefficients for incidence from medium 1 on medium 2 (similarly ρ_{21} , ρ_{23} , τ_{21}), and $\mathcal{L}$ is the propagation factor across the layer.

- Use the identity (Taylor series)

$$\sum_{p=0}^{\infty} \chi^p = \frac{1}{1 - \chi} \quad (29)$$

$$\Rightarrow \rho_{\text{eff}} = \rho_{12} + \frac{(1 + \rho_{12})(1 - \rho_{12})\rho_{23}\mathcal{L}^2}{1 + \rho_{12}\rho_{23}\mathcal{L}^2} \quad (30)$$

$$\Rightarrow \boxed{\rho_{\text{eff}} = \frac{\rho_{12} + \rho_{23} e^{-2\gamma_2 d \cos \bar{\theta}_2}}{1 + \rho_{12}\rho_{23} e^{-2\gamma_2 d \cos \bar{\theta}_2}}} \quad (31)$$

the same result as Eq. (22).

3.2 Implications

1. γ_2 is complex valued, so for $d \gtrsim \lambda_2$ (the wavelength in medium 2), $|\rho_{\text{eff}}|$ should have oscillatory behavior, with maxima corresponding to constructive interference of the multiple reflections and minima $\rightarrow$ destructive interference.
2. For $d \gg \delta_s$, $\rho_{\text{eff}} \rightarrow \rho_{12}$.
3. For $\rho_{23} \ll 1$, $\rho_{\text{eff}} \rightarrow \rho_{12}$.

Variables

Units are SI. Values are given for universal constants; “typical” values are representative, not definitive.

Symbol	Description	Units	Value
<i>Constants and wavenumber</i>			
j	imaginary unit, $j^2 = -1$ (time dependence $e^{j\omega t}$)	–	–
k_0	free-space angular wavenumber	rad/m	–
<i>Constitutive parameters</i>			
$\varepsilon_1, \varepsilon_2, \varepsilon_3$	complex dielectric constants (relative) of media 1, 2, 3, $\varepsilon' - j\varepsilon''$; real parts $\varepsilon'_1, \varepsilon'_2, \varepsilon'_3$	dimensionless	–
$\varepsilon'_1, \varepsilon'_2$	relative permittivities of media 1 and 2	dimensionless	≥ 1 (almost always)
μ_1, μ_2	permeabilities of media 1 and 2	H/m	μ_0 in RS
<i>Boundary conditions and geometric optics</i>			
$\mathbf{E}_1, \mathbf{E}_2$	(total) electric field in media 1 and 2	V/m	–
$\mathbf{H}_1, \mathbf{H}_2$	(total) magnetic field in media 1 and 2	A/m	–
$E^\perp, H^\perp$	field components normal to the interface	V/m; A/m	–
$\mathbf{E}^\parallel, \mathbf{H}^\parallel$	field components tangential to the interface	V/m; A/m	–
$\hat{\mathbf{a}}$	unit normal to the interface	dimensionless	–
θ_i, θ_r	angles of incidence and reflection	rad	$\theta_i = \theta_r = \theta_1$
<i>Layered media</i>			
d	thickness of the layer (medium 2)	m	–
N	number of layers in a composite	dimensionless	3 here
x, y, z	Cartesian coordinates	m	interfaces at $z = 0, -d$
$\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$	Cartesian unit vectors	dimensionless	–

Symbol	Description	Units	Value
$\theta_1, \tilde{\theta}_2, \tilde{\theta}_3$	angles of propagation (from the normal) in media 1, 2, 3	rad	complex in lossy media (real if lossless)
$\gamma_1, \gamma_2, \gamma_3$	propagation constants of media 1, 2, 3, $jk_0\sqrt{\epsilon}$	m^{-1}	–
η_1, η_2, η_3	intrinsic impedances of media 1, 2, 3	Ω	–
$\hat{\mathbf{k}}$	propagation unit vector	dimensionless	–
$\tilde{\mathbf{E}}_1^-, \tilde{\mathbf{E}}_1^+$	downward- and upward-propagating $\mathbf{E}$ phasors in medium 1 (likewise medium 2)	V/m	–
$\tilde{\mathbf{E}}_1, \tilde{\mathbf{E}}_2, \tilde{\mathbf{E}}_3$	total $\mathbf{E}$ phasors in media 1, 2, 3	V/m	–
$\tilde{\mathbf{H}}_1, \tilde{\mathbf{H}}_2, \tilde{\mathbf{H}}_3$	total $\mathbf{H}$ phasors in media 1, 2, 3	A/m	–
$\tilde{H}_{1x}, \tilde{H}_{1z}, \dots$	x and z components of $\tilde{\mathbf{H}}_1$ (likewise media 2, 3)	A/m	–
E_h^i	amplitude of the incident H-polarized field (Lecture 5), here A_1	V/m	–
A_1, A_2, A_3	amplitudes of the downward-propagating $\mathbf{E}$ fields in media 1, 2, 3	V/m	A_1 known
B_1, B_2	amplitudes of the upward-propagating $\mathbf{E}$ fields in media 1 and 2	V/m	–
ρ_{eff}	effective reflection coefficient of the composite, B_1/A_1	dimensionless	ρ_{12} if $d \gg \delta_s$
Γ	reflectivity, $ \rho_{\text{eff}} ^2$	dimensionless	0–1
ρ_{12}, ρ_{23}	Fresnel reflection coefficients for incidence from medium 1 on 2 and from 2 on 3 (H or V pol.)	dimensionless	–
ρ_{21}	Fresnel reflection coefficient for incidence from medium 2 on 1	dimensionless	$-\rho_{12}$
τ_{12}, τ_{21}	Fresnel transmission coefficients from medium 1 to 2 and from 2 to 1	dimensionless	H pol.: $1 + \rho_{12}, 1 - \rho_{12}$
$\mathcal{L}$	propagation factor across the layer, $e^{-\gamma_2 d \cos \tilde{\theta}_2}$	dimensionless	$ \mathcal{L} \leq 1$
χ	ratio of successive terms, $\rho_{21}\rho_{23}\mathcal{L}^2$	dimensionless	$ \chi < 1$ (medium 1 air, $\epsilon' \geq 1, \theta_1 < 90^\circ$)
p	summation index	dimensionless	0, 1, 2, ...
λ_2	wavelength in medium 2	m	–
α, α_2	attenuation constant (of medium 2), $\text{Re}(\gamma)$	Np/m	0 if lossless
δ_s	skin depth, $1/\alpha$	m	–
δ_p	penetration depth, $1/(2\alpha)$	m	$\delta_s/2$